

1. Let p , q and r be statements. Use the laws of logical equivalence to show that

$$p \rightarrow (q \vee r) \equiv (p \wedge \sim q) \rightarrow r.$$

(3 marks)

$$\begin{aligned} p \rightarrow (q \vee r) &\equiv \sim p \vee (q \vee r) && \text{by def. of } \rightarrow \\ &\equiv (\sim p \vee q) \vee r && \text{by associative law} \\ &\equiv (\sim p \vee \sim(\sim q)) \vee r && \text{by double negative law} \\ &\equiv \sim(p \wedge \sim q) \vee r && \text{by de Morgan's law} \\ &\equiv (p \wedge \sim q) \rightarrow r && \text{by def. of } \rightarrow \end{aligned}$$

2. For each of the following statements, write down the negation of the statement, and state which of the original or the negation is true. Justify your answers.

(a) $\exists x \in \mathbb{Z}^+$ such that $\forall y \in \mathbb{Z}^+, x \geq y$.

(2 marks)

Negation: $\forall x \in \mathbb{Z}^+, \exists y \in \mathbb{Z}^+$ such that $x < y$.

The negation is true. For any $x \in \mathbb{Z}^+$, choose $y = x + 1$. Then $x < y$.

(b) $\forall r, s \in \mathbb{R}$, if $rs \in \mathbb{Q}$, then $r \in \mathbb{Q}$ and $s \in \mathbb{Q}$.

(2 marks)

Negation: $\exists r, s \in \mathbb{R}$ such that $rs \in \mathbb{Q}$ and $r \notin \mathbb{Q}$ or $s \notin \mathbb{Q}$.

The negation is true. Let $r = \sqrt{2}$ and $s = \sqrt{2}$.
Then $r, s \notin \mathbb{Q}$ and $rs \in \mathbb{Q}$.

3. For each of the following statements, either prove the statement or give a counterexample to disprove the statement.

(5 marks)

(a) For all $a, b, c \in \mathbb{Z}^+$, if $a \mid b$ and $b \mid c$, then $a \mid (b + c)$.

Proof Suppose $a, b, c \in \mathbb{Z}^+$ and $a \mid b$ and $b \mid c$.

Then $b = ak$ for some integer k ,
and $c = bl$ for some integer l .

$$\begin{aligned} \text{Hence } b + c &= ak + bl \\ &= ak + akl \\ &= a(k + kl) \end{aligned}$$

Since $k + kl \in \mathbb{Z}$, we have $a \mid (b + c)$.

(b) For all $a, b, c \in \mathbb{Z}^+$, if $a \mid (b + c)$, then $a \mid b$ or $a \mid c$.

False. Consider $a = 2$ $b = 3$ and $c = 5$.

Then $a \mid (b + c)$ since $2 \mid 8$
but $a \nmid b$ and $a \nmid c$ since $2 \nmid 3$ and $2 \nmid 5$.

4. Use mathematical induction to prove that $3^n + 5 < 4^n$ for each integer $n \geq 2$.

(5 marks)

Let $P(n)$ be the predicate $3^n + 5 < 4^n$.

Consider $n=2$.

$3^2 + 5 = 14$ and $4^2 = 16$. Since $14 < 16$, $P(2)$ is true.

Suppose $k \in \mathbb{Z}$, $k \geq 2$ and $P(k)$ is true. Hence $3^k + 5 < 4^k$.

Consider $P(k+1)$.

$$\begin{aligned}
 3^{k+1} + 5 &= 3 \cdot 3^k + 5 \\
 &< 3 \cdot 3^k + 15 && \text{since } 5 < 15 \\
 &= 3(3^k + 5) \\
 &< 3 \cdot 4^k && \text{by ind. hypothesis} \\
 &< 4 \cdot 4^k && \text{since } 3 < 4 \text{ and } 4^k > 0 \\
 &= 4^{k+1}
 \end{aligned}$$

Hence $3^{k+1} + 5 < 4^{k+1}$.

Thus, by mathematical induction, $3^n + 5 < 4^n$ $\forall$ integers $n \geq 2$.

5. (a) Use the Euclidean algorithm to determine $\gcd(931, 301)$. Show your working.

(2 marks)

$$931 = 301 \cdot 3 + 28$$

$$301 = 28 \cdot 10 + 21$$

$$28 = 21 \cdot 1 + 7$$

$$21 = 7 \cdot 3 + 0$$

$$\therefore \gcd(931, 301) = 7$$

- (b) Write down the unique prime factorisation of 931 in standard form.

(1 mark)

$$\begin{aligned} 931 &= 7 \cdot 133 \\ &= 7 \cdot 7 \cdot 19 \\ &= 7^2 \cdot 19 \end{aligned}$$

- (c) How many positive divisors does 931 have? (Hint: use part (b)).

(1 mark)

Can choose $0, 1, 2$ 7's
 $0, 1$ 19's
 $\therefore 6$ positive divisors

6. Recall that $\emptyset$ denotes the empty set. Let $A = \{1, 2\}$, $B = \{-1, 2\}$, and $C = \{\emptyset, \{-1\}\}$.

Write down the following sets, by listing their elements and using appropriate brackets.

(5 marks)

(a) $A \cup B = \{-1, 1, 2\}$

(b) $B \cap C = \emptyset$

(c) $\mathcal{P}(C) = \{\emptyset, \{\emptyset\}, \{\{-1\}\}, \{\emptyset, \{-1\}\}\}$

(d) $(A \times B) - (B \times B) = \{(1, -1), (1, 2)\}$

$$A \times B = \{(1, -1), (1, 2), (2, -1), (2, 2)\}$$

$$B \times B = \{(-1, -1), (-1, 2), (2, -1), (2, 2)\}$$

(e) $\{X \mid (X \subseteq A) \wedge (X \in C)\} = \{\emptyset\}$

7. (a) Consider the function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ where for each element $(m, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ we define

$$f((m, n)) = 2^m 3^n.$$

(i) Determine $f((3, 1))$.

(1 mark)

$$2^3 \cdot 3^1 = 24$$

(ii) Determine the pre-image of $\{36\}$.

(2 marks)

$$\text{Since } 36 = 2^2 \cdot 3^2 \\ f^{-1}(\{36\}) = \{(2, 2)\}$$

(iii) Prove that the function f is one-to-one.

(4 marks)

Suppose $(a, b), (m, n) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ and

$$f((a, b)) = f((m, n))$$

$$2^a \cdot 3^b = 2^m \cdot 3^n$$

$$2^{a-m} = 3^{n-b}$$

The only power of 2 equal to a power of 3 is $2^0 = 3^0$ so $a-m=0$ and $n-b=0$.

Thus $a=m$ and $n=b$ so

$$(a, b) = (m, n).$$

$\therefore f$ is one-to-one.

Note: could use unique prime factorisation instead.

(b) Let A and B be sets. Which one of the following statements is implied by the existence of a one-to-one function $g : A \rightarrow B$? (circle your answer)

(1 mark)

$$|A| \geq |B|$$

$$|A| \leq |B|$$

$$|A| = |B|$$

$$|A| \neq |B|$$

8. Let ρ be the relation defined on $\mathbb{Z}^+$ as follows. For all $a, b \in \mathbb{Z}^+$, $a\rho b$ if and only if

$$a^2 - b^2 \equiv 0 \pmod{5}.$$

(a) Prove that ρ is an equivalence relation.

(5 marks)

$\forall a \in \mathbb{Z}^+, a^2 - a^2 \equiv 0 \pmod{5}$ so $a\rho a$.
Thus ρ is reflexive.

Suppose $a, b \in \mathbb{Z}^+$ and $a\rho b$.

$$\text{Thus } a^2 - b^2 \equiv 0 \pmod{5}$$

$$b^2 - a^2 \equiv -1(a^2 - b^2) \equiv -1(0) \equiv 0 \pmod{5}$$

Thus ρ is symmetric.

Suppose $a, b, c \in \mathbb{Z}^+$ and $a\rho b$ and $b\rho c$.

$$\text{Thus } a^2 - b^2 \equiv 0 \pmod{5} \text{ and } b^2 - c^2 \equiv 0 \pmod{5}.$$

$$\text{Hence } a^2 - c^2 \equiv (a^2 - b^2) + (b^2 - c^2) \equiv 0 + 0 \equiv 0 \pmod{5}.$$

Thus ρ is transitive.

Since ρ is reflexive, symmetric and transitive, it is an equivalence relation.

(b) How many equivalence classes does the relation ρ have?

3

(1 mark)

(c) List the elements of the equivalence class $[1]$ that are greater than 1 and less than 10.

4, 6, 9

(1 mark)

9. The Cayley table for a group $(G, *)$ is given below. (You do not need to prove that this is a group.)

$*$	a	b	c	d	e	f
a	a	b	c	d	e	f
b	b	a	f	e	d	c
c	c	e	a	f	b	d
d	d	f	e	a	c	b
e	e	c	d	b	f	a
f	f	d	b	c	a	e

(a) Is this group abelian? Justify your answer.

No. $b * c = f$ but $c * b = e$. (1 mark)

(b) Which element is the identity of this group? Justify your answer.

a is the identity (1 mark)
since $a * x = x * a = x$ for all $x \in G$.

(c) Which element is the inverse of the element b ? Justify your answer.

$b^{-1} = b$ since $b * b = a$ (1 mark)

(d) Write down the elements of the subgroup $\langle e \rangle$.

$\langle e \rangle = \{e, f, a\}$ (1 mark)

Note: ok to just write e, f, a without brackets.

10. Solve the equation $9x - 7 = 4$ in the field $(\mathbb{Z}_{23}, +, \cdot)$, and hence determine the smallest non-negative integer y such that $9y - 7 \equiv 4 \pmod{23}$. *Hint:* $9 \cdot 18 \equiv 1 \pmod{23}$.

(3 marks)

$$9x - 7 = 4$$

$$9x = 11$$

$$18 \cdot 9 \cdot x = 18 \cdot 11$$

$$x = 198$$

$$\text{Since } 198 \equiv 14 \pmod{23}$$

$$y = 14.$$

11. Suppose that Alice, Bob, Charlie, Delia, Edward, Fred and George are organising a dinner party and each person will bring one dish to share.

- (a) For the dinner party, two people will bring desserts and the other five will bring savoury dishes. How many different choices are there for the two people who will bring desserts?

$$\binom{7}{2} = 21$$

(1 mark)

- (b) At the party, the seven friends stand in a line for a photograph. How many different arrangements are there of the seven friends standing in a line?

$$7! = 5040$$

(1 mark)

- (c) How many different arrangements are there of the seven friends standing in a line such that Alice is not next to Charlie and Bob is not next to Delia.

ways with Alice next to Charlie

(4 marks)

$$6! \times 2$$

ways with Bob next to Delia

$$6! \times 2$$

ways with Alice next to Charlie and Bob next to Delia

$$5! \times 4$$

ways with Alice not next to Charlie and Bob not next to Delia.

$$7! - (6! \times 2 + 6! \times 2 - 5! \times 4)$$

$$= 5040 - 2400$$

$$= 2640$$

12. Let S be the set of integers from 1 to 99, so $S = \{x \in \mathbb{Z} : 1 \leq x \leq 99\}$. How many integers must you pick from S in order to be sure that at least one of them is divisible by 3? Justify your answer.

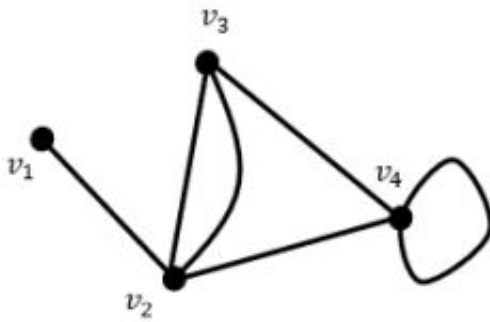
(3 marks)

Consider the partition of S given by $\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}, \dots, \{97, 98, 99\}$.

We can choose 2 from each part (the first two) to get 66 integers, none of which is divisible by 3. But if we choose 67 then by the pigeonhole principle, we must have 3 from one part and hence we are guaranteed to have at least one integer that is divisible by 3.

Note: There are other ways to justify this.

13. Let G be the graph shown below.



- (a) Write down an adjacency matrix for G . Make sure you clearly label the rows and columns of your matrix.

(2 marks)

$$\begin{array}{c}
 v_1 \quad v_2 \quad v_3 \quad v_4 \\
 \begin{array}{c}
 v_1 \\
 v_2 \\
 v_3 \\
 v_4
 \end{array}
 \begin{bmatrix}
 0 & 1 & 0 & 0 \\
 1 & 0 & 2 & 1 \\
 0 & 2 & 0 & 1 \\
 0 & 1 & 1 & 1
 \end{bmatrix}
 \end{array}$$

- (b) Does G have an Euler circuit? If yes, write it out as a sequence of vertices. If not, determine the fewest number of edges that would need to be added to G to create a new graph that does have an Euler circuit.

(2 marks)

No, but if you add the edge $\{v_1, v_3\}$ then you would have an Euler circuit

14. A tree has 11 vertices, of which exactly four vertices have degree 1 and exactly one vertex has degree 4. Determine the degrees of the remaining six vertices. Show your working.

(4 marks)

Tree with 11 vertices has 10 edges.

so $\sum \text{degrees} = 20$.

Let $d_1, d_2, d_3, d_4, d_5, d_6$ be the degrees of the remaining vertices.

By the handshake theorem

$$1 + 1 + 1 + 1 + 4 + d_1 + d_2 + d_3 + d_4 + d_5 + d_6 = 20$$

$$\sum_{i=1}^6 d_i = 12$$

since each d_i must be ≥ 1 , we have

$$d_i = 2 \quad \forall i = 1, \dots, 6.$$

END OF EXAMINATION